3030AFE: Assignment 1 (2026)

Assignment Guide (read carefully!)

- Answer all THREE (3) questions.
- Total marks: 25. Marking is based on code, outputs, quality of answers and aesthetics.
- All answers must be in **Jupyter Notebook**. `COLAB` and `VS-CODE` are not accepted. Do not copy outputs in `MS-WORD` for comments.
- Present your `codes` & `outcomes` for all questions.
- Type up your `comments` with **Markdown** [Question 1 (part 4), Question 2 (all) & Question 3 (parts 1,2,3,5)].
- Save your answers in two files: **ipynb** and **pdf** and submit both files online.
- The submitted files should contain your `name`, `student ID`, `Python codes`, `outcomes`, and `comments`.
- Use codes learned from the course -- **Use of AI is not permitted, justification is required if you use codes beyond those taught in the course.**
- Due: **Thursday, 27 August 2026**.
- Late submission is subject to substantial penalty.

In [34]:
```
# essential libraries
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
import sklearn
%matplotlib inline
sns.set()

# libraries for clustering & scaling
from sklearn.cluster import KMeans
from sklearn.cluster import AgglomerativeClustering as AC
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler

# Use scipy.cluster.hierarchy to draw dendrogram
import scipy.cluster.hierarchy as shc
```

Question 1 (7 marks)

Use the dataset `Risk_Profile_26.csv` to answer the following questions. The dataset contains information on 121 fictional Australian firms, with 7 columns: `State`, `Firm Type`, `Firm`, `Management Quality`, `Operational Risk`, `Compliance Score`, and `Revenue Growth`.

1. Rename the columns to `State`, `Type`, `Firm`, `Quality`, `Risk`, `Score` and `Growth`.
2. Create dummy variables for `State` and, using these dummy variables, answer the following questions.
   - How many firms are in each state?
   - How many states have at most 15 firms?
   - What percentage of firms are in `Queensland`, and how many in `South Australia`?
3. How many firms have a `Score` between 4 and 6 (inclusive)?
4. For each state, find the firm with the highest `Quality` score (i.e., the highest level of management quality).
5. Split the `Type` column into `Firm_Region` and `Firm_Size`. Create a boxplot of `Risk` for each `Firm_Size` category. Briefly comment on the plot.
6. Create a new column `Score_Cat` with three categories:
   - `Low` (`Score <= 5`)
   - `Mid` (`5 < Score <= 7`)
   - `High` (`Score > 7`)

Make a bar plot showing the number of firms in each category.

In [ ]:
```
# load data
R=pd.read_csv('Risk_Profile_26.csv')
R.info()
```

Question 2 (11 marks)

Use the dataset `Risk_Profile_26.csv` with renamed columns to answer this question.

1. Use `StandardScaler` to scale `Risk`, `Score`, and `Growth`. Discuss whether scaling should be applied to these variables prior to clustering.
2. Find the optimal number of clusters when KMeans clustering is applied to the scaled `Risk`, `Score` and `Growth`.
3. Use scaled `Risk` and `Growth` to form 4 clusters using KMeans (`set the random_state of KMeans with the last 4 digits of your student ID`). Make a scatter plot on `Risk` (x-axis) and `Growth` (y-axis) with centroids, assigning a distinct color for each cluster. Comment on the results.
4. Find the optimal number of clusters when agglomerative clustering (AC) is applied on the scaled `Risk` and `Growth`.
5. Use scaled `Risk` and `Growth` to conduct AC with 4 clusters. Make a scatter plot of `Risk` (x-axis) and `Growth` (y-axis), assigning a distinct color for each cluster. Comment on the results.
6. Compare clusters formed by KMeans and AC. Comment on their similarities and differences.

In [ ]:
```
# load data
R=pd.read_csv("Risk_Profile_26.csv")
R.info()
```

Question 3 (7 marks)

Use data in `GDP_25.xlsx` and `Health_25.xlsx` to answer this question. Each file contains five columns: `Country`, `Country Code`, `1960`, `2025`, and `Region`. The `GDP_25` file reports `GDP` for `1960` and `2025`, while the `Health_25` file reports `Life Expectancy` for the same years.

1. Calculate the median and interquartile range of `Life Expectancy` in `1960` and `2025`. Compare the results.
2. Create a violinplot of `GDP` for `1960` and `2025`. Use the plot to comment on the `GDP` distributions.
3. Merge the two datasets into a single dataframe. Then, calculate the correlation coefficient between `GDP` and `Life Expectancy` by `Region` for `1960` and `2025` separately. Comment on the results.
4. List the top 5 countries with the largest increase in `GDP`. Calculate the median change in `GDP` (from `1960` to `2025`) by `Region`.
5. . Create two scatterplots showing the relationship betwe `GDP` and `Life Expectancy` in 1960 and 2025. Display the plots in the same row using subplots and use different colours for the Regions. Comment on any differences in the relationship across the two years..

In [ ]:
```
GDP = pd.read_excel('GDP_25.xlsx')
Health = pd.read_excel('Health_25.xlsx')
GDP.head()
Health.head()
```

Assignment 1 - Predictive Analytics

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In [4]:
```
# essential libraries
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
import sklearn
%matplotlib inline
sns.set()

# libraries for clustering & scaling
from sklearn.cluster import KMeans
from sklearn.cluster import AgglomerativeClustering as AC
from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import StandardScaler

# Use scipy.cluster.hierarchy to draw dendrogram
import scipy.cluster.hierarchy as shc
```

Question 1

In [8]:
```
rp = pd.read_csv("Risk_Profile_26.csv")
```

In [9]:
```
rp.head()
```

Out[9]:

	State	Firm Type	Firm	Management Quality	Operational Risk	Compliance Score	Revenue Growth
0	New South Wales	New South Wales, Medium	NSW Firm 001	37	1.865	4.605	3.045
1	Queensland	Queensland, Medium	QLD Firm 001	36	2.293	4.486	2.553
2	South Australia	South Australia, Medium	SA Firm 001	47	2.054	5.161	-3.036
3	Victoria	Victoria, Medium	VIC Firm 001	42	2.357	4.844	6.033
4	Tasmania	Tasmania, Large	TAS Firm 001	81	1.486	8.425	1.765

In [10]:
```
#1
rp.columns = ['State','Type','Firm','Quality','Risk','Score','Growth']
rp.head(3)
```

Out[10]:

	State	Type	Firm	Quality	Risk	Score	Growth
0	New South Wales	New South Wales, Medium	NSW Firm 001	37	1.865	4.605	3.045
1	Queensland	Queensland, Medium	QLD Firm 001	36	2.293	4.486	2.553
2	South Australia	South Australia, Medium	SA Firm 001	47	2.054	5.161	-3.036

In [11]:
```
#2
pd.get_dummies(rp['State']).head()
```

Out[11]:

	New South Wales	Queensland	South Australia	Tasmania	Victoria	Western Australia
0	True	False	False	False	False	False
1	False	True	False	False	False	False
2	False	False	True	False	False	False
3	False	False	False	False	True	False
4	False	False	False	True	False	False

In [12]:
```
#2.1
rp.State.value_counts()
```

Out[12]:
```
State
New South Wales      36
Victoria             32
Queensland           30
Western Australia    12
South Australia       9
Tasmania              2
Name: count, dtype: int64
```

In [13]:
```
#2.2
rp.State.value_counts()<=15
```

Out[13]:
```
State
New South Wales      False
Victoria             False
Queensland           False
Western Australia     True
South Australia       True
Tasmania              True
Name: count, dtype: bool
```

In [14]:
```
#2.3
rp.State.value_counts(normalize=True)*100
```

Out[14]:
```
State
New South Wales      29.752066
Victoria             26.446281
Queensland           24.793388
Western Australia     9.917355
South Australia       7.438017
Tasmania              1.652893
Name: proportion, dtype: float64
```

In [15]:
```
#3
rp[(rp.Score >=4) & (rp.Score <=6)]
```

Out[15]:

	State	Type	Firm	Quality	Risk	Score	Growth
0	New South Wales	New South Wales, Medium	NSW Firm 001	37	1.865	4.605	3.045
1	Queensland	Queensland, Medium	QLD Firm 001	36	2.293	4.486	2.553
2	South Australia	South Australia, Medium	SA Firm 001	47	2.054	5.161	-3.036
3	Victoria	Victoria, Medium	VIC Firm 001	42	2.357	4.844	6.033
6	Victoria	Victoria, Medium	VIC Firm 002	33	2.463	5.186	2.772
...	...	...	...	...	...	...	...
110	Victoria	Victoria, Medium	VIC Firm 029	39	3.095	5.309	0.258
111	Queensland	Queensland, Small	QLD Firm 028	30	2.291	5.145	6.254
112	New South Wales	New South Wales, Medium	NSW Firm 035	32	2.956	4.438	3.068
117	Victoria	Victoria, Medium	VIC Firm 031	41	1.948	5.155	6.573
119	Queensland	Queensland, Small	QLD Firm 029	38	1.884	4.645	2.089

61 rows × 7 columns

In [16]:
```
#4
i=rp.groupby('State').Quality.idxmax()
rp.iloc[i,:][['State','Firm','Quality']].sort_values(by='Quality', ascending=False)
```

Out[16]:

	State	Firm	Quality
28	New South Wales	NSW Firm 008	90
76	Tasmania	TAS Firm 002	90
96	Victoria	VIC Firm 025	89
18	Western Australia	WA Firm 001	79
116	South Australia	SA Firm 009	74
13	Queensland	QLD Firm 003	62

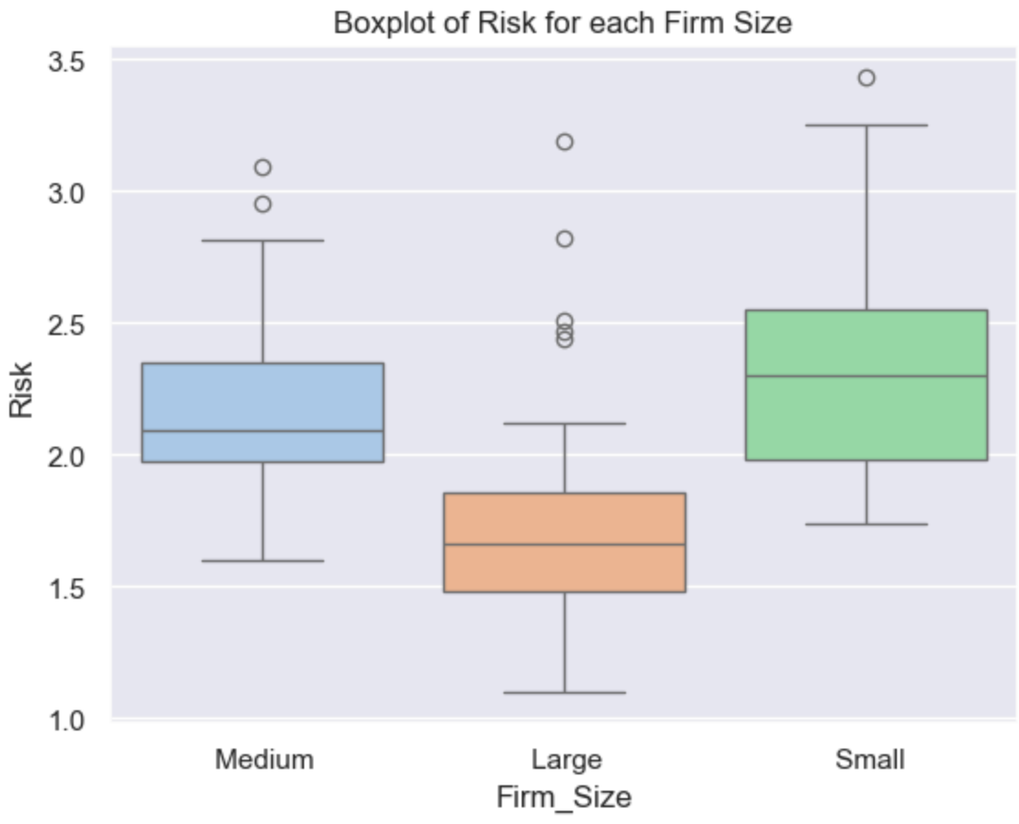
In [41]:
```
#5
rp[['Firm_Region', 'Firm_Size']] = rp['Type'].str.split(',', expand=True)
rp.head()
```

Out[41]:

	State	Type	Firm	Quality	Risk	Score	Growth	Firm_Region	Firm_Size	Score_Ca
0	New South Wales	New South Wales, Medium	NSW Firm 001	37	1.865	4.605	3.045	New South Wales	Medium	Lo
1	Queensland	Queensland, Medium	QLD Firm 001	36	2.293	4.486	2.553	Queensland	Medium	Lo
2	South Australia	South Australia, Medium	SA Firm 001	47	2.054	5.161	-3.036	South Australia	Medium	Mi
3	Victoria	Victoria, Medium	VIC Firm 001	42	2.357	4.844	6.033	Victoria	Medium	Lo
4	Tasmania	Tasmania, Large	TAS Firm 001	81	1.486	8.425	1.765	Tasmania	Large	Hig

In [43]:
```
#5
sns.boxplot(x="Firm_Size", y="Risk", data=rp, hue = "Firm_Size", palette='pastel')
plt.title("Boxplot of Risk for each Firm Size")
```

Out[43]:
```
Text(0.5, 1.0, 'Boxplot of Risk for each Firm Size')
```



5. Comment:

The boxplot compares the distribution of Risk values across three firm size categories: Medium, Large, and Small. Overall, risk levels vary notably depending on the scale of the firm, with Large firms demonstrating lower risk levels compared to Medium and Small firms.

Small Firms show the highest overall risk, featuring both the highest median (around 2.3) and the largest interquartile range (IQR). Their risk values spread widely, reaching up to around 3.25, with an extreme outlier near 3.4.

Medium Firms exhibit a moderate risk profile, with a median risk level near 2.1. While the central 50% of the data remains relatively tightly clustered between 2.0 and 2.4, there are a couple of high-value outliers approaching 3.0 to 3.1.

Large Firms display the lowest central risk level, centered around a median of approximately 1.7. However, this category contains several upper outliers ranging from around 2.4 up to nearly 3.2, indicating that while most large firms maintain low risk, a few exceptional cases experience significantly elevated risk levels.

In [19]:
```
rp.Score.idxmax()
```

Out[19]:
```
35
```

In [20]:
```
#6
rp['Score_Cat'] = pd.cut(rp['Score'], bins=[0,5,7,35], labels=['Low','Mid','High'])
rp.head(5)
```

Out[20]:

	State	Type	Firm	Quality	Risk	Score	Growth	Firm_Region	Firm_Size	Score_Ca
0	New South Wales	New South Wales, Medium	NSW Firm 001	37	1.865	4.605	3.045	New South Wales	Medium	Lo
1	Queensland	Queensland, Medium	QLD Firm 001	36	2.293	4.486	2.553	Queensland	Medium	Lo
2	South Australia	South Australia, Medium	SA Firm 001	47	2.054	5.161	-3.036	South Australia	Medium	Mi
3	Victoria	Victoria, Medium	VIC Firm 001	42	2.357	4.844	6.033	Victoria	Medium	Lo
4	Tasmania	Tasmania, Large	TAS Firm 001	81	1.486	8.425	1.765	Tasmania	Large	Hig

In [44]:
```
#6
sns.countplot(x='Score_Cat',data=rp, hue = 'Score_Cat', palette='rainbow')
plt.title("Number of firms in each score category")
plt.xlabel("Score category")
plt.ylabel("Number of firms")
```

Out[44]:
```
Text(0, 0.5, 'Number of firms')
```



Question 2

In [23]:
```
#1
scaler = StandardScaler()
rp1 = rp.loc[:,["Risk", "Score", "Growth"]]
rpscale= scaler.fit_transform(rp1)
rpscale=pd.DataFrame(rpscale)
rpscale.describe()
```

Out[23]:

	0	1	2
count	1.210000e+02	1.210000e+02	1.210000e+02
mean	-3.413248e-16	-4.881311e-16	-8.808381e-17
std	1.004158e+00	1.004158e+00	1.004158e+00
min	-2.072078e+00	-2.253311e+00	-4.746601e+00
25%	-7.319477e-01	-7.309577e-01	-5.395809e-01
50%	-1.399486e-01	-2.514419e-01	-1.404602e-02
75%	5.778233e-01	5.574676e-01	5.353768e-01
max	2.989190e+00	2.166542e+00	2.007031e+00

1. Comment:

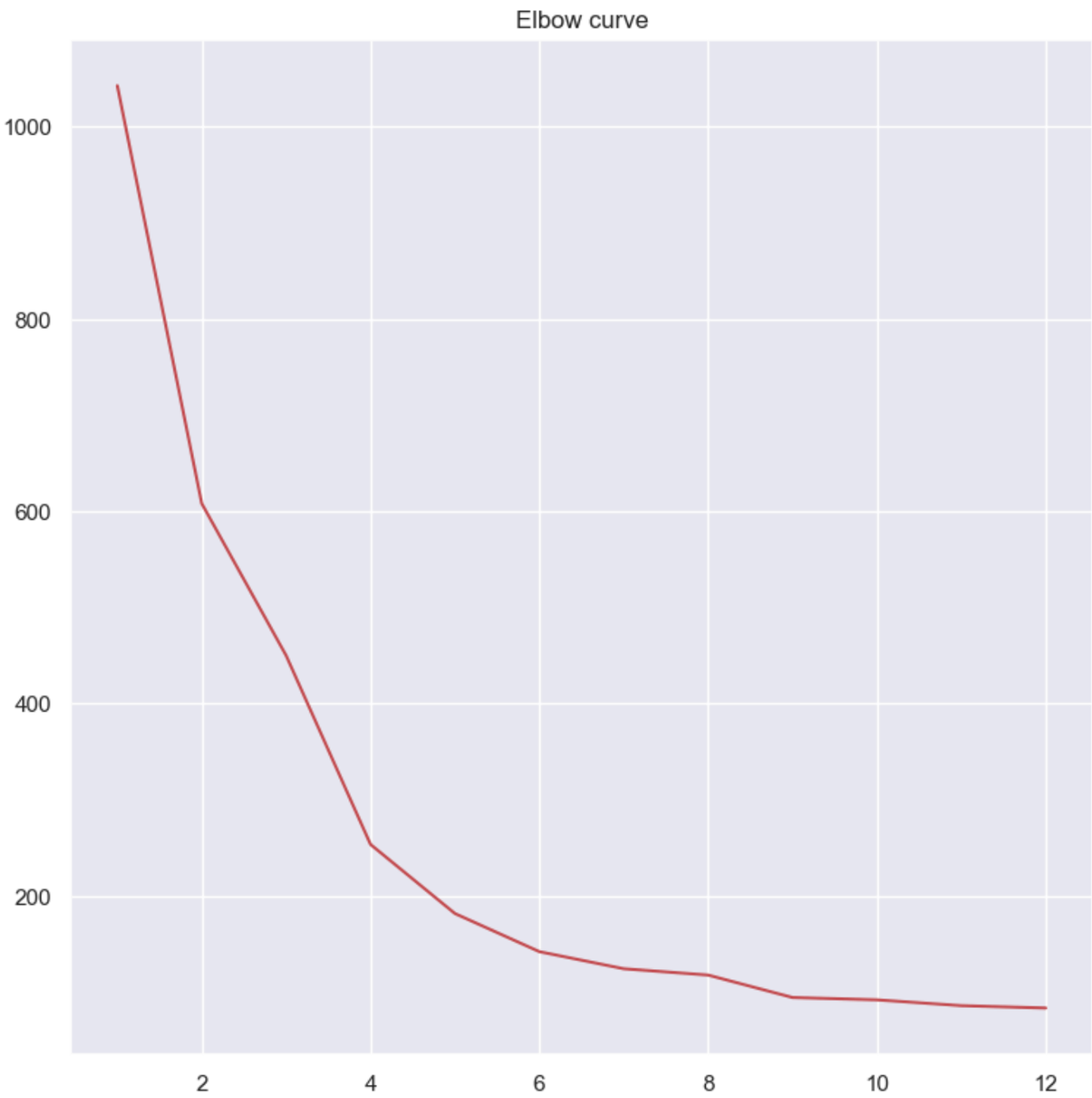
Scaling should be applied prior to distance-based clustering algorithms like K-Means or Hierarchical Clustering. As shown in the table, all features now have a mean of approximately 0 and a standard deviation of approximately 1. Without scaling, variables measured on larger scales or with higher variance would disproportionately influence the distance calculation, causing the algorithm to bias clusters toward those specific variables. Standardization ensures every feature contributes equally to the similarity measurements.

In [24]:
```
#2
# define a function called Elbow_Curve
# there are two inputs for the function -- X (data) and kmax (maximum number of clusters
def Elbow_Curve(x, kmax):
 # inertia_: Sum of Squared Distances
 inertia = []

 # no. of clusters
 for k in range(1, kmax+1):
  km = KMeans(n_clusters=k)
  km=km.fit(x)
  inertia.append(km.inertia_)
```

```
 # make a line plot of inertia with plt
 fig = plt.figure(figsize=(9, 9))
 #lineplot: plt.plot(x,y,'r-') with red line
 plt.plot(range(1, kmax+1), inertia, 'r-')
 plt.title('Elbow curve')
```

In [25]:
```
#2
Elbow_Curve(rp1,12)
```

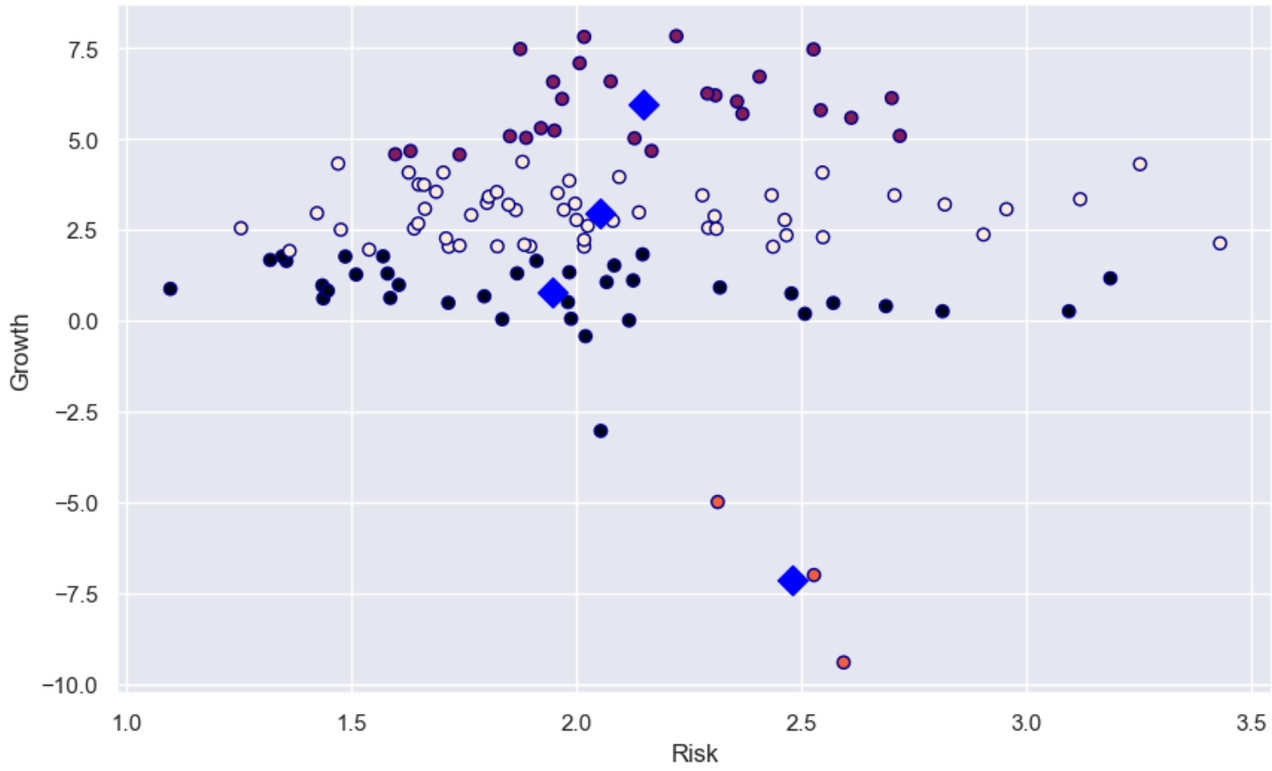


2. Comment:

The optimal number of clusters is k = 4 (or k = 6). The primary elbow occurs at k = 4, where the steep drop in inertia levels off into a noticeably gentler slope. A secondary inflection point appears at k = 6, after which the curve completely flattens out, indicating diminishing returns for adding more clusters. Thus, k = 4 provides the most optimal balance between cluster compactness and model simplicity.

In [26]:
```
#3
rp3 = rp.loc[:,["Risk","Growth"]]
# cluster size is set to be 4
km = KMeans(n_clusters = 4, random_state=1867)
km.fit(rp3)
km_centers = km.cluster_centers_
# ploting the graph of the clusters
plt.figure(figsize=(10,6))
plt.scatter(x= rp3.iloc[:, 0], y=rp3.iloc[:, 1],c= km.labels_, edgecolor='darkblue')
plt.scatter(x= km_centers[:,0], y=km_centers[:,1], c='blue', marker='D',s=100)
plt.xlabel('Risk')
plt.ylabel('Growth')
```

Out[26]:
```
Text(0, 0.5, 'Growth')
```



3. Comment:

The topmost cluster (purple points) represents a high growth, moderate risk group. Centered around a Growth score of approximately 6.0 and a Risk score near 2.15, this segment comprises high-performing firms that achieve substantial growth while incurring average risk levels.

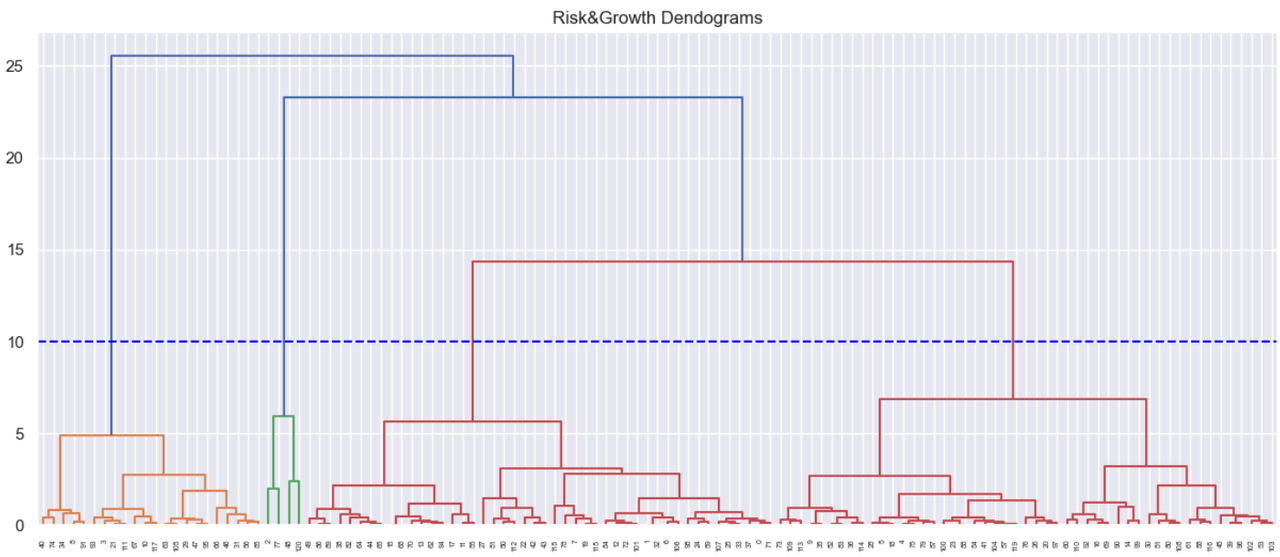
The middle cluster (white points) captures a moderate growth, moderate-to-high risk segment. Centered near a Growth score of 3.0 and Risk of 2.05, it forms the largest cluster, spanning a wide horizontal range of risk levels (from ~1.5 to over 3.4) while maintaining moderate positive growth.

The lower-central cluster (dark navy points) indicates a low growth, broad risk profile. With its centroid situated around a Growth score of 0.8 and Risk near 1.95, these firms exhibit minimal to near-zero growth across a broad spectrum of low to high risk values.

The bottom-right cluster (red points) isolates a small group of severe negative growth, high risk outliers. Centered around a Growth score near -7.0 and a Risk score of 2.48, this cluster clearly separates firms experiencing severe contraction combined with elevated risk from the rest of the dataset.

In [27]:
```
#4
# dendogram plot
plt.figure(figsize=(15, 6))
plt.title("Risk&Growth Dendograms")
# method = "ward" & metric="euclidean"
dend = shc.dendrogram(shc.linkage(rp3, method="ward", metric="euclidean"))
# plot a blue dash line (height: y=180)
plt.axhline(y = 10, color = 'blue', linestyle = '--')
```

Out[27]:
```
<matplotlib.lines.Line2D at 0x170cc1450>
```



4. Comment:

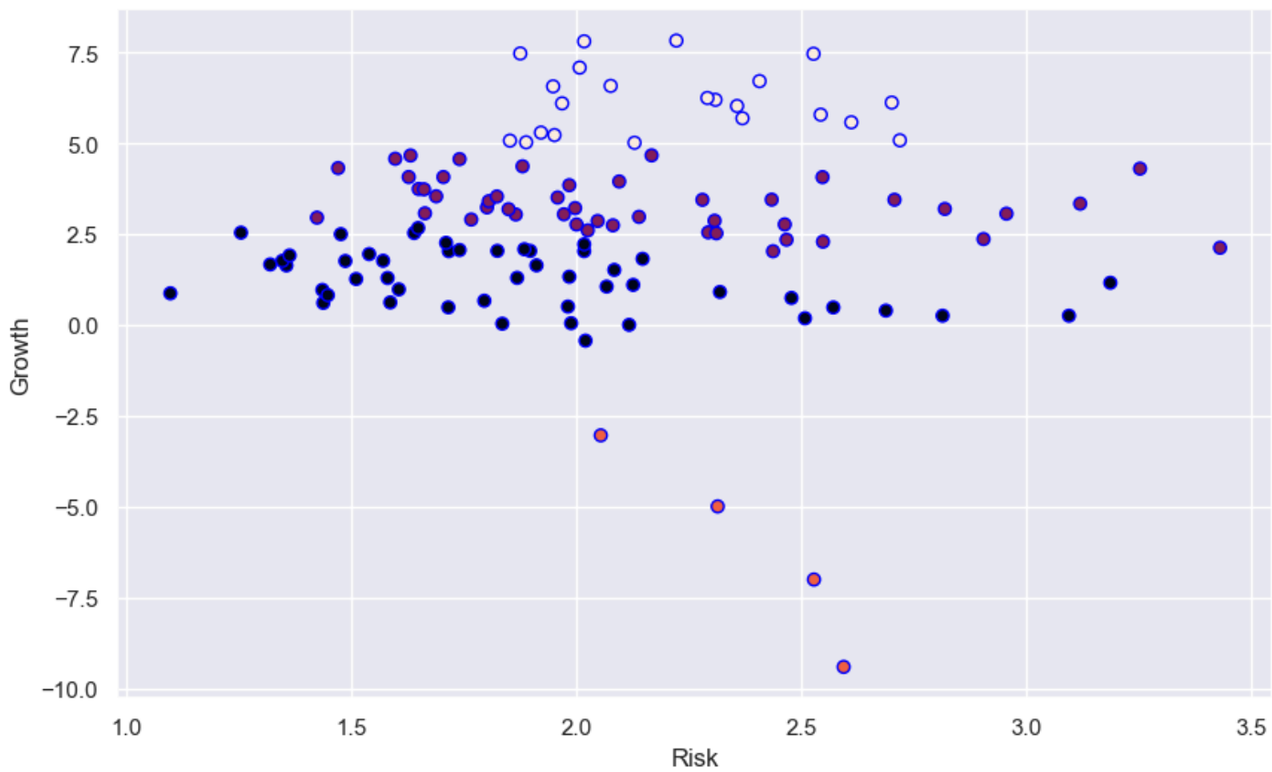
Based on the dendrogram, the optimal number of clusters for Agglomerative Clustering is 4. There is a big drop in the linkage distance around height 10, where the blue dashed horizontal line cuts through four vertical lines. This indicates that partitioning the data into 4 distinct clusters provides the best balance between maximizing between-cluster separation and preserving within-cluster cohesion.

In [28]:
```
#5
# form 4 clusters with AgglomerativeClustering (AC)
ac = AC(n_clusters=4)
# use fit_predict to get cluster labels
ac.fit_predict(rp3)
```

Out[28]:
```
array([1, 1, 2, 3, 0, 0, 1, 1, 3, 0, 3, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0, 3,
       1, 0, 1, 1, 0, 1, 0, 3, 0, 3, 1, 1, 3, 0, 0, 1, 1, 0, 3, 0, 1, 1,
       1, 0, 3, 3, 2, 1, 1, 0, 0, 0, 0, 1, 3, 0, 0, 1, 0, 0, 1, 3, 1, 1,
       3, 3, 1, 0, 1, 1, 1, 0, 3, 0, 0, 2, 1, 0, 0, 1, 1, 0, 1, 3, 1, 0,
       0, 1, 0, 3, 0, 3, 1, 3, 0, 0, 1, 0, 0, 1, 0, 0, 0, 3, 1, 1, 0, 0,
       0, 3, 1, 0, 0, 1, 0, 3, 1, 0, 2])
```

In [45]:
```
#5
# plot the clusters
plt.figure(figsize=(10,6))
plt.scatter(x= rp3.iloc[:, 0], y=rp3.iloc[:, 1], c=ac.fit_predict(rp3),edgecolor='blue')
plt.xlabel('Risk')
plt.ylabel('Growth')
```

Out[45]:
```
Text(0, 0.5, 'Growth')
```



5. Comment:

High growth cluster (white points) gathers firms achieving high growth rates ranging approximately from 5 to nearly 8, spanning moderate risk levels around 1.8 to 2.8.

Moderate growth cluster (purple points) captures firms with moderate growth rates mostly between 2.5 and 5, extending across a wider range of risk levels from about 1.4 to 3.4.

Low growth cluster (blue points) represents the largest group, containing firms concentrated around lower growth rates between 0 and 2.5 across a wide risk spectrum ranging from 1.1 to 3.2.

Severe negative growth cluster (red points) isolated a small set of distinct outlier firms experiencing significant negative growth (from around -3 down to -9.5) alongside elevated risk levels above 2.0.

6. Comment:

Both K-Means and Agglomerative Clustering (AC) successfully partition the dataset into four overall risk-growth profiles, exhibiting strong similarities. Both algorithms identify the isolated cluster of extreme negative growth outliers at the bottom and separate the remaining data into low, moderate,

and high growth tiers along the vertical axis. Additionally, both methods show that Risk spans broadly across most growth levels rather than driving clear vertical splits. The main difference lies in how the two algorithms define the boundaries between the upper growth tiers. K-Means forms more spherical, distance-to-centroid-based clusters, resulting in a large central group of moderate growth that extends across the entire horizontal range of risk values (~1.5 to 3.4). In contrast, Agglomerative Clustering uses hierarchical link distances, which creates a strictly high growth cluster at the top and places the broad middle band into a single moderate growth group. Consequently, AC produces cleaner horizontal boundary cuts based on Growth thresholds, whereas K-Means clusters are shaped by spatial proximity around central medoids.

Question 3

In [30]:
```
gdp = pd.read_excel("GDP_25.xlsx")
health = pd.read_excel("Health_25.xlsx")
```

In [31]:
```
gdp.head()
```

Out[31]:

	Country	Country_Code	1960	2025	Region
0	Afghanistan	AFG	NaN	413.757895	Asia
1	Albania	ALB	NaN	8575.171346	Europe
2	Algeria	DZA	238.392477	5364.027950	Africa
3	American Samoa	ASM	NaN	NaN	Oceania
4	Andorra	AND	NaN	46812.448449	Europe

In [32]:
```
health.head()
```

Out[32]:

	Country	Country_Code	1960	2025	Region
0	Afghanistan	AFG	32.799	66.035	Asia
1	Albania	ALB	56.413	79.602	Europe
2	Algeria	DZA	40.839	76.261	Africa
3	American Samoa	ASM	65.053	72.852	Oceania
4	Andorra	AND	72.094	84.041	Europe

In [33]:
```
#1
health.describe()
```

Out[33]:

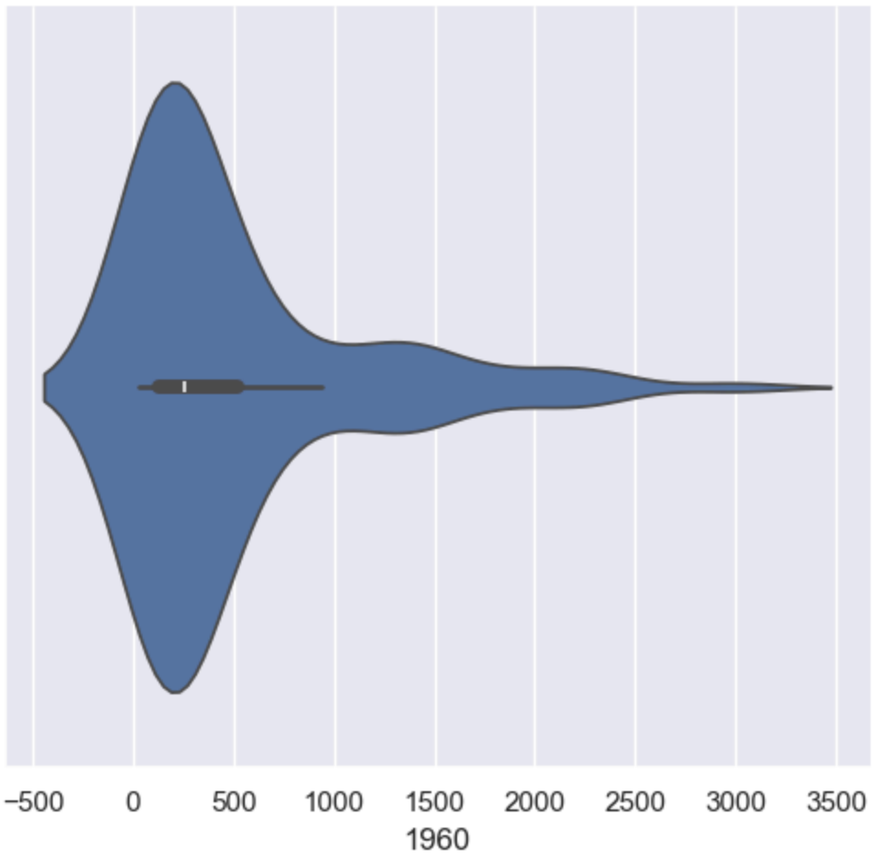
	1960	2025
count	209.000000	211.000000
mean	54.570008	73.664343
std	11.658058	7.058775
min	28.054000	54.462000
25%	44.436000	68.410500
50%	55.153000	74.183780
75%	64.242000	78.749000
max	74.040073	86.372000

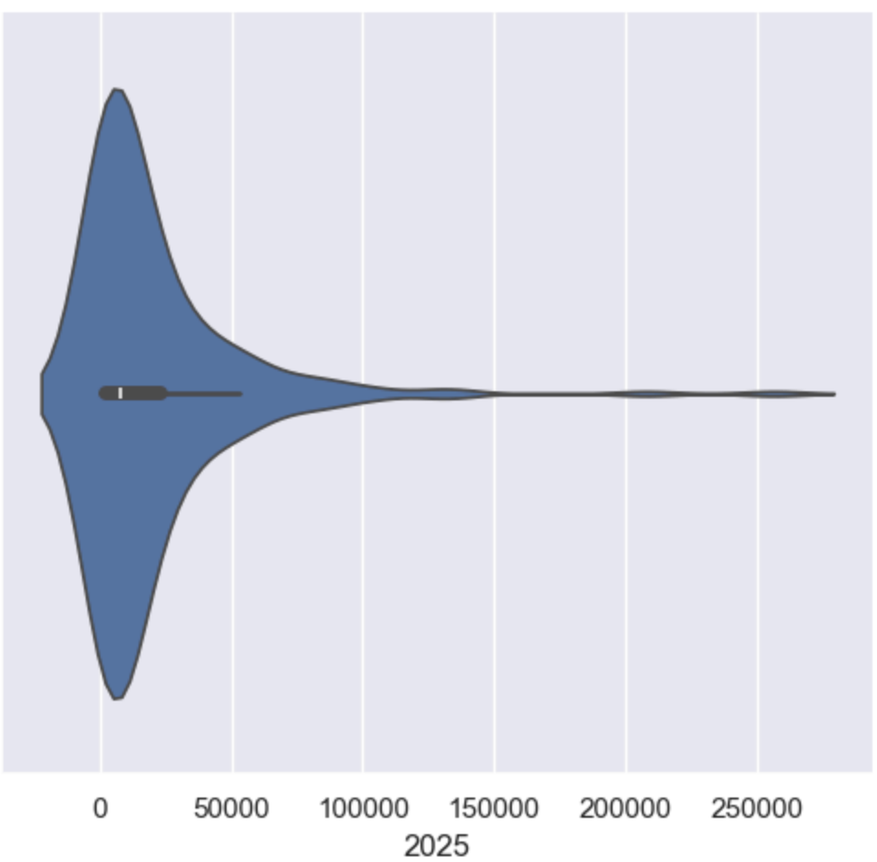
1. Comment:

Between 1960 and 2025, the overall life expectancy increased significantly while global variability decreased. In 1960, the median life expectancy stood at approximately 55.15 years, with an Interquartile Range of 19.81 years. By 2025, the median life expectancy rose by nearly 19 years to 74.18 years, while the IQR narrowed to 10.34 years. This comparison highlights both a substantial improvement in central overall life expectancy and a tighter grouping around higher ages, reflecting reduced global inequality in life expectancy over time.

In [34]:
```
#2
sns.catplot(x= 1960, data=gdp, kind='violin')
sns.catplot(x= 2025, data=gdp, kind='violin')
```

Out[34]: <seaborn.axisgrid.FacetGrid at 0x30658f610>





2. Comment:

In 1960, the GDP distribution was tightly clustered at lower values, with the primary density bulked heavily between 0 and 500. The distribution exhibits a mild right tail extending up to around 3,500, indicating that while a few entities had higher economic output, global GDP levels were relatively compact and contained within a narrow absolute range.

By 2025, the overall scale of GDP expanded exponentially, accompanied by extreme right-skewness. The main body of the distribution remains heavily concentrated near lower values (mostly under 25,000 to 50,000), but the right tail stretches dramatically out past 250,000. This extremely long, thin tail reflects immense economic growth among top-tier entities, highlighting a sharp increase in global economic inequality and variance compared to 1960.

In [35]:
```
#3
mergetable = pd.merge(gdp,health,on=('Country','Country_Code','Region'), how='outer')
mergetable.rename(columns = {'1960_x':'GDP_1960', '2025_x':'GDP_2025','1960_y':'LifeExp_
mergetable
```

Out[35]:

	Country	Country_Code	GDP_1960	GDP_2025	Region	LifeExp_1960	LifeExp_2025
0	Afghanistan	AFG	NaN	413.757895	Asia	32.799000	66.035000
1	Albania	ALB	NaN	8575.171346	Europe	56.413000	79.602000
2	Algeria	DZA	238.392477	5364.027950	Africa	40.839000	76.261000
3	American Samoa	ASM	NaN	NaN	Oceania	65.053000	72.852000
4	Andorra	AND	NaN	46812.448449	Europe	72.094000	84.041000
...	...	...	...	...	...	...	...
206	Virgin Islands (U.S.)	VIR	NaN	NaN	Americas	62.743537	80.519512
207	West Bank and Gaza	PSE	NaN	3455.028529	Asia	NaN	65.170000
208	Yemen, Rep.	YEM	NaN	426.354076	Asia	33.462000	69.295000
209	Zambia	ZMB	221.559849	1330.727806	Africa	50.648000	66.349000
210	Zimbabwe	ZWE	276.419784	2156.034093	Africa	53.492000	62.775000

211 rows × 7 columns

In [46]:
```
#3
#compute Pearson correlation between two columns by region
mergetable.groupby('Region')[['GDP_1960','LifeExp_1960']].corr()
```

Out[46]:

		GDP_1960	LifeExp_1960
Region			
Africa	**GDP_1960**	1.000000	0.307853
	LifeExp_1960	0.307853	1.000000
Americas	**GDP_1960**	1.000000	0.643158
	LifeExp_1960	0.643158	1.000000
Asia	**GDP_1960**	1.000000	0.544012
	LifeExp_1960	0.544012	1.000000
Europe	**GDP_1960**	1.000000	0.545412
	LifeExp_1960	0.545412	1.000000
Oceania	**GDP_1960**	1.000000	0.911188
	LifeExp_1960	0.911188	1.000000

In [47]:
```
#3
#compute Pearson correlation between two columns by region
mergetable.groupby('Region')[['GDP_2025','LifeExp_2025']].corr()
```

Out[47]:

Region		GDP_2025	LifeExp_2025
Africa	**GDP_2025**	1.000000	0.566486
	LifeExp_2025	0.566486	1.000000
Americas	**GDP_2025**	1.000000	0.529807
	LifeExp_2025	0.529807	1.000000
Asia	**GDP_2025**	1.000000	0.752395
	LifeExp_2025	0.752395	1.000000
Europe	**GDP_2025**	1.000000	0.666643
	LifeExp_2025	0.666643	1.000000
Oceania	**GDP_2025**	1.000000	0.770261
	LifeExp_2025	0.770261	1.000000

3. Comment:

The correlation analysis across regions reveals notable shifts in the strength of the linear relationship between GDP and life expectancy from 1960 to 2025. Overall, every region maintains a positive relationship between economic output and life expectancy across both periods, though the degree of coupling varies significantly by geography and time.

In 1960, the relationship was strongest in Oceania (r = 0.911), indicating a near-perfect linear association between higher GDP and longer life expectancy. The Americas (r = 0.643), Europe (r = 0.545), and Asia (r = 0.544) displayed moderate positive correlations. Conversely, Africa exhibited the weakest correlation (r = 0.308), suggesting that economic output played a relatively minor role in explaining variations in life expectancy across African nations at that time.

By 2025, the regional landscape experienced several key shifts. Asia (r = 0.752) and Africa (r = 0.566) saw significant increases in correlation strength compared to 1960, indicating that growth in GDP became increasingly tied to gains in life expectancy in these developing regions. Europe also showed a moderate rise to r = 0.667. Conversely, Oceania saw its correlation weaken to r = 0.770, and the Americas decreased slightly to r = 0.530, pointing toward diminishing marginal returns on life expectancy as economic output reaches advanced levels.

In [38]:
```
#4
mergetable['GDP_Increase']= mergetable.GDP_2025 - mergetable.GDP_1960
mergetable.sort_values(by='GDP_Increase', ascending=False).head(5)
```

Out[38]:

	Country	Country_Code	GDP_1960	GDP_2025	Region	LifeExp_1960	LifeExp_202
22	Bermuda	BMU	1715.314632	132604.439086	Americas	66.482000	82.30900
112	Luxembourg	LUX	2261.177418	131407.681365	Europe	68.800561	83.35853
89	Ireland	IRL	706.551022	103887.800388	Europe	70.194268	82.86097
183	Switzerland	CHE	1954.311321	100631.802755	Europe	71.313415	84.05609
143	Norway	NOR	1451.285078	87497.328930	Europe	73.549756	83.11219

In [48]:
```
#4
mergetable[['GDP_Increase','Region']].groupby('Region').describe()
```

Out[48]:

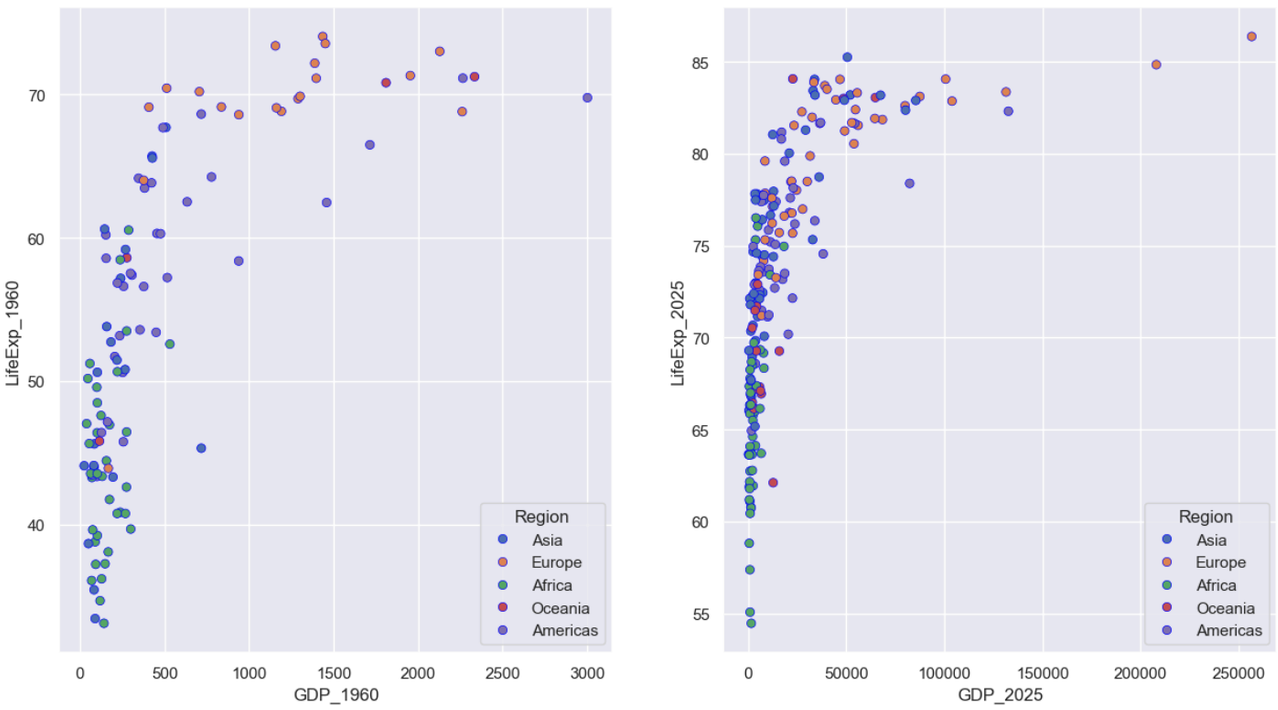
Region	count	mean	std	min	25%	50%	
Africa	35.0	2840.177831	3641.244777	121.169205	841.435377	1397.608758	3448.5
Americas	31.0	20315.514712	26138.159071	1634.955639	6351.561046	13409.231019	21276.7
Asia	21.0	18039.445690	22300.708204	663.847654	2468.536461	7092.286738	32963.0
Europe	19.0	58793.085189	29625.174859	12254.764425	40808.460814	53399.808201	72794.3
Oceania	4.0	29451.956751	29931.770235	2850.537814	4921.113307	25966.037822	50496.8

In [40]:
```
#5
X1960 = mergetable.loc[:, ['GDP_1960', 'LifeExp_1960', 'Region']]
X2025 = mergetable.loc[:, ['GDP_2025', 'LifeExp_2025', 'Region']]

fig, (ax1, ax2) = plt.subplots(nrows=1, ncols=2, figsize=(15, 8))

sns.scatterplot(data=X1960, x='GDP_1960', y='LifeExp_1960', hue='Region', edgecolor='blu
sns.scatterplot(data=X2025, x='GDP_2025', y='LifeExp_2025', hue='Region', edgecolor='blu
```

Out[40]: `<Axes: xlabel='GDP_2025', ylabel='LifeExp_2025'>`



5. Comment:

The two scatter plots reveal several significant changes in the relationship between GDP and life expectancy from 1960 to 2025.

The entire range of life expectancy shifts substantially upward. In 1960, life expectancy spans from under 35 years to around 74 years, whereas by 2025, the minimum life expectancy rises past 54 years, with the upper boundary reaching nearly 87 years. This reflects widespread global improvements in healthcare and standard of living over the 65-year period.

The horizontal GDP axis undergoes an exponential scale expansion. In 1960, GDP values are heavily concentrated below 500, capping out around 3,000. By 2025, overall economic output grows massively, stretching the x-axis past 250,000. Despite this scale expansion, the Preston curve remains intact, where initial increases in GDP correspond to steep gains in life expectancy before flattening out at higher wealth levels.

In 1960, regions are more sharply segregated along the curve: African nations cluster heavily at the bottom with low GDP and life expectancy below 50 years, while European nations occupy the upper tier above 68 years. By 2025, there is clear regional convergence: African and Asian nations show marked improvements in life expectancy, moving up into the 60–75 range, while European and high-income Asian nations converge along the flattened upper plateau above 80 years.